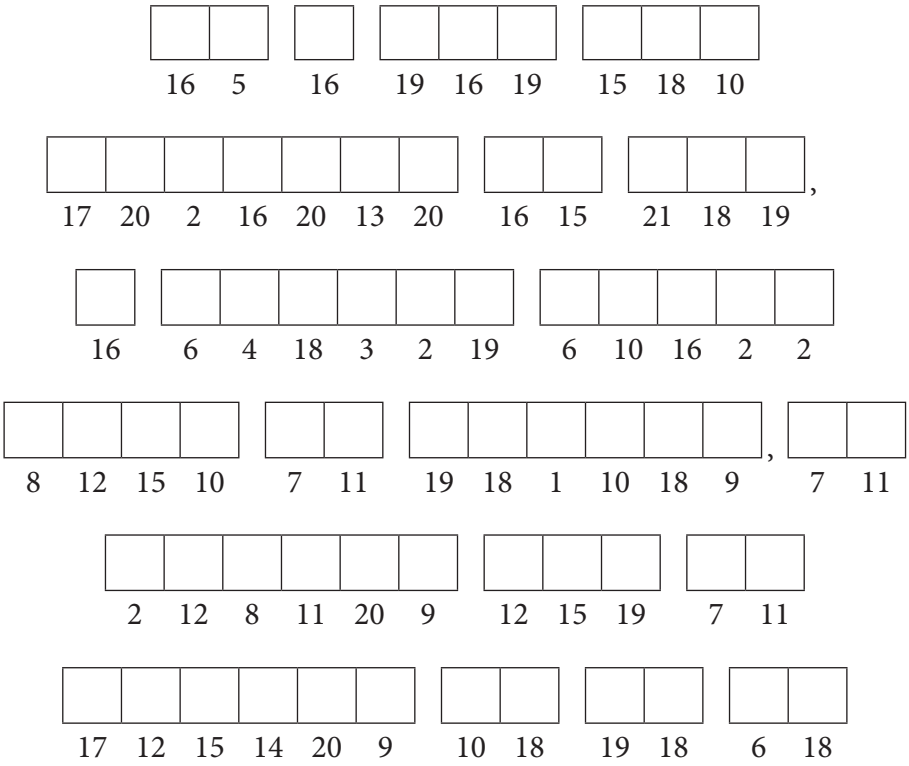
Name: _____

WORKSHEET

Review of index laws



G. K. Chesterton

Rules

$a^m \times a^n = a^{m+n}$ $\quad$ $a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ $\quad$ $a^0 = 1$

$$a^{-n} = \frac{1}{a^n}$$

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Questions

Simplify the following.

1 $x^3 \times x^5 =$

2 $y^8 \div y^2 =$

3 $(x^3)^4 =$

4 $y^5 \times y^2 \times y^7 =$

5 $x^{10} \div x^2 \div x^4 =$

6 $(y^5)^3 \times (y^2)^4 =$

7 $3x^8 \times 4x^5 \div 2x^7 =$

8 $(y^3)^7 \div y^{14} \times y^2 =$

9 $x^2y^5 \times x^4y^7 =$

10 $15x^9y^3 \div 3x^4y =$

11 $(x^2y^3)^5 \div x^4y^5 \times xy^2 =$

12 $20x^7y^4 \div 5x^3y^2 \times (2x^2y)^2 =$

13 $x^9y^{15} \div (x^3y^5)^3 =$

14 $\dfrac{3x^6y}{y^3} \times \dfrac{4x^7y^5}{x^4} =$

15 $\dfrac{5x^2y^5}{x^5y^7} \times \dfrac{4x^7y}{10x^3y^2} =$

16 $\dfrac{(2x^3)^4}{x^9y} \times \dfrac{x^5y^7}{8xy^4} =$

17 $\dfrac{15x^3y^7}{(x^2y)^5} \times \dfrac{2x^4y^5}{3x^6y^3} =$

18 $\dfrac{3x^2y}{2x^6y^7} \times \dfrac{4x^5y^4}{9x^3y^2} =$

19 $\dfrac{4x^7y^9}{xy^2} \div \dfrac{(2x^3y^5)^2}{3x^2y^4} =$

20 $\dfrac{5x^3y^4}{2x^8y} \times \dfrac{8x^7y^5}{10x^6y^2} =$

21 $\dfrac{3x^7y^{12}}{(x^5y^3)^2} \times \dfrac{5x^4y^3}{4xy^7} \div \dfrac{(2x^2y^4)^3}{x^8} =$

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Solutions

A $16x^8y^4$

B $\frac{10y^4}{x^9}$

C x^8

D $3x^2y$

E $\frac{2y^6}{x^4}$

F x^4

G $\frac{15x^2}{32y^{10}}$

H y^{14}

I $2x^7y^2$

K $12x^9y^3$

L y^6

M $6x^6$

N $\frac{2x}{y^3}$

O $\frac{2}{3x^2y^4}$

R x^6y^{12}

S y^{23}

T $5x^5y^2$

U x^{12}

V 1

W y^9

Y x^7y^{12}

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